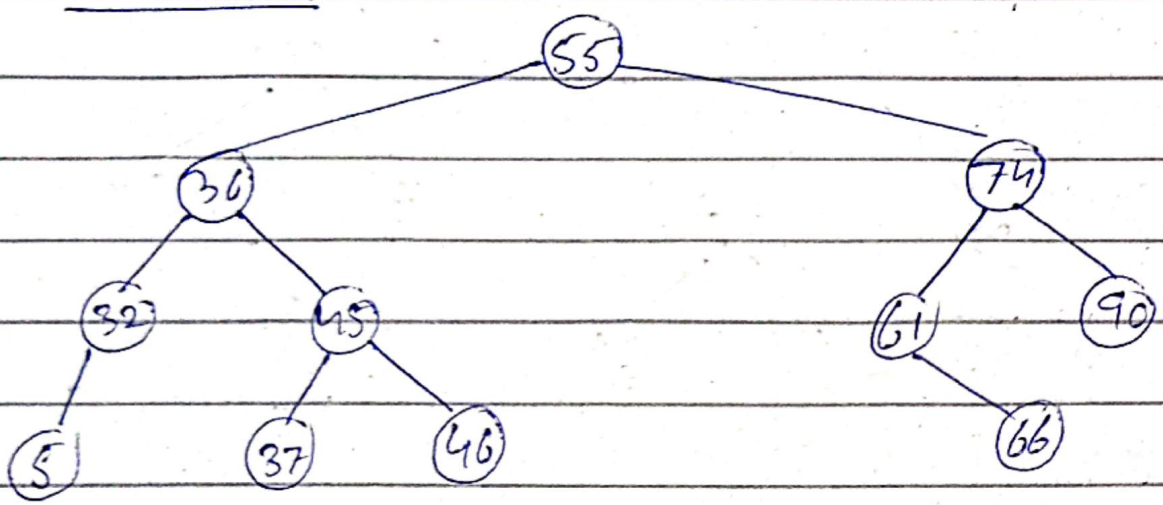
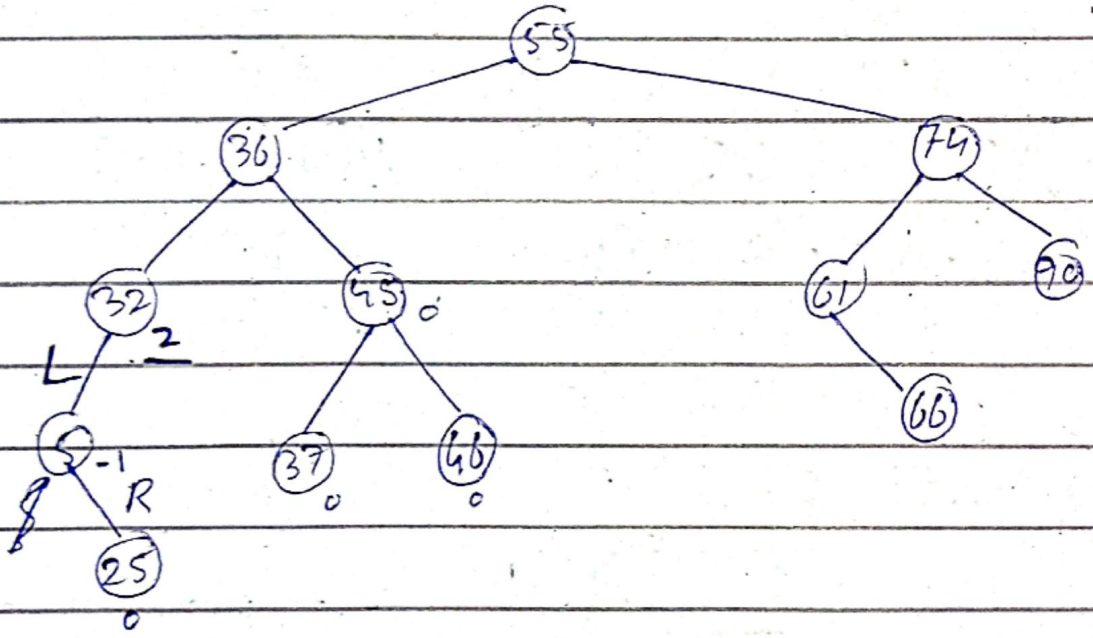
Asad Ali 25L-0519 BCS-3G

Data Structures 3 - 1

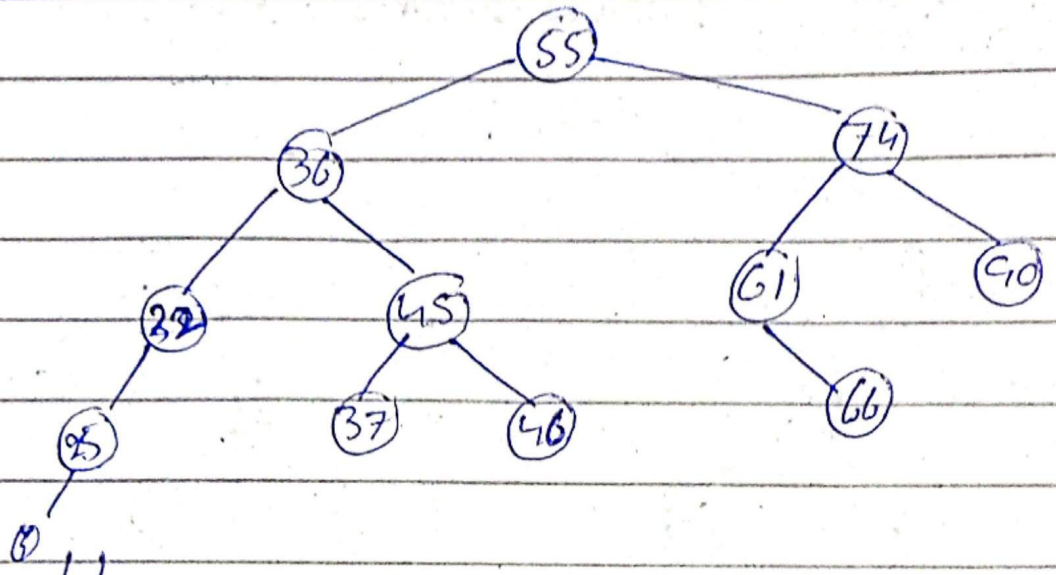
Initial Tree :-



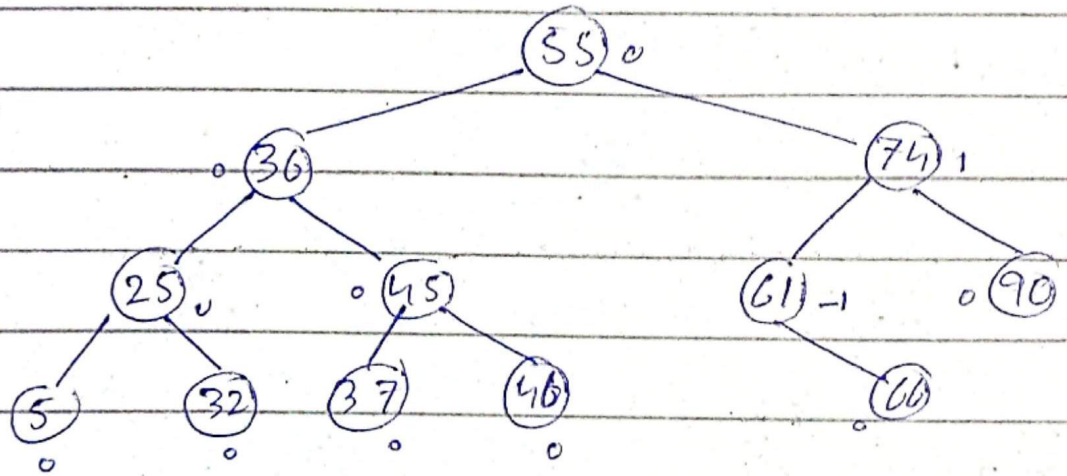
i- insert 25 :-



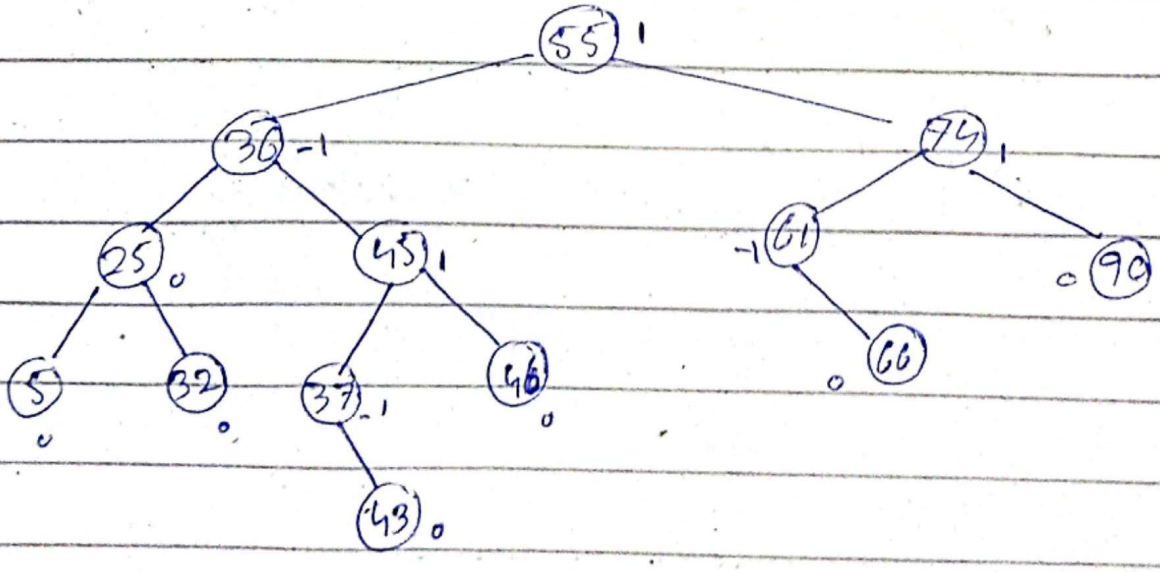
LR $\xrightarrow{rot}$ LL ⟶ balanced



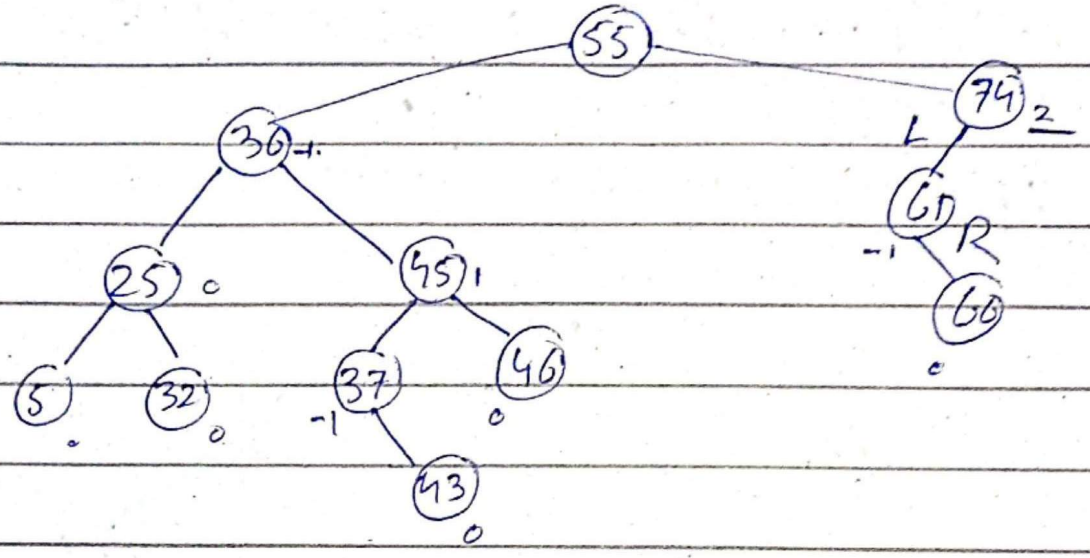
LL



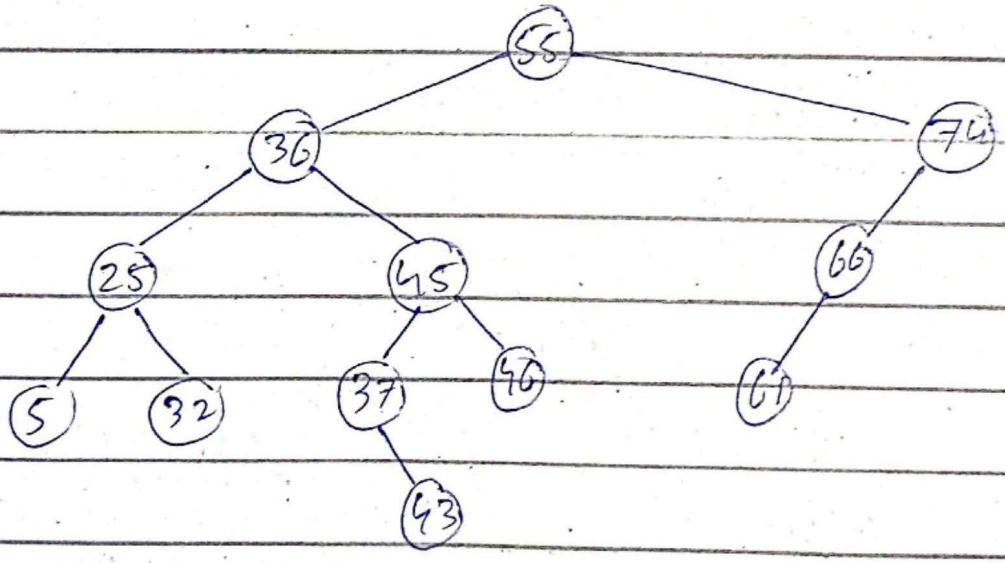
ii- insert 43 :-



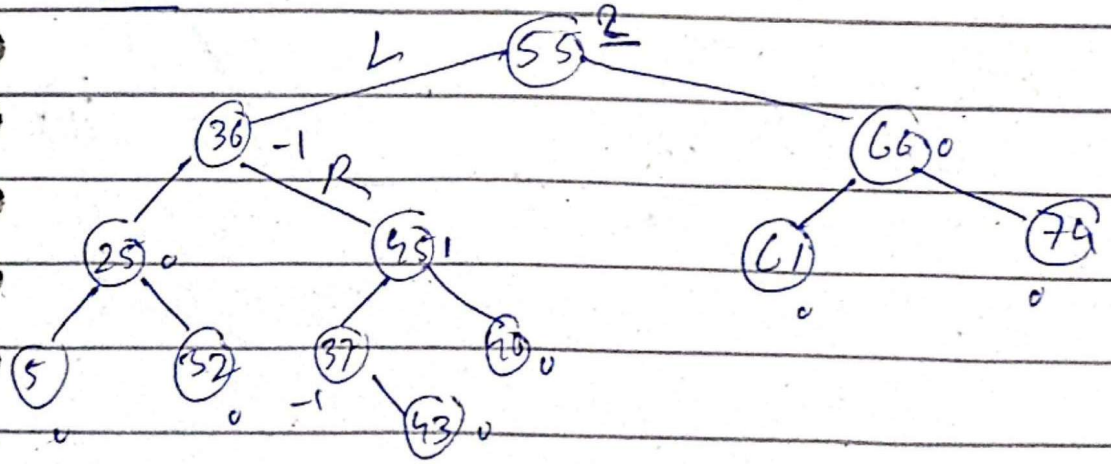
iii- delete 90:-



LR $\xrightarrow{rot}$ LL $\xrightarrow{rot}$ balanced

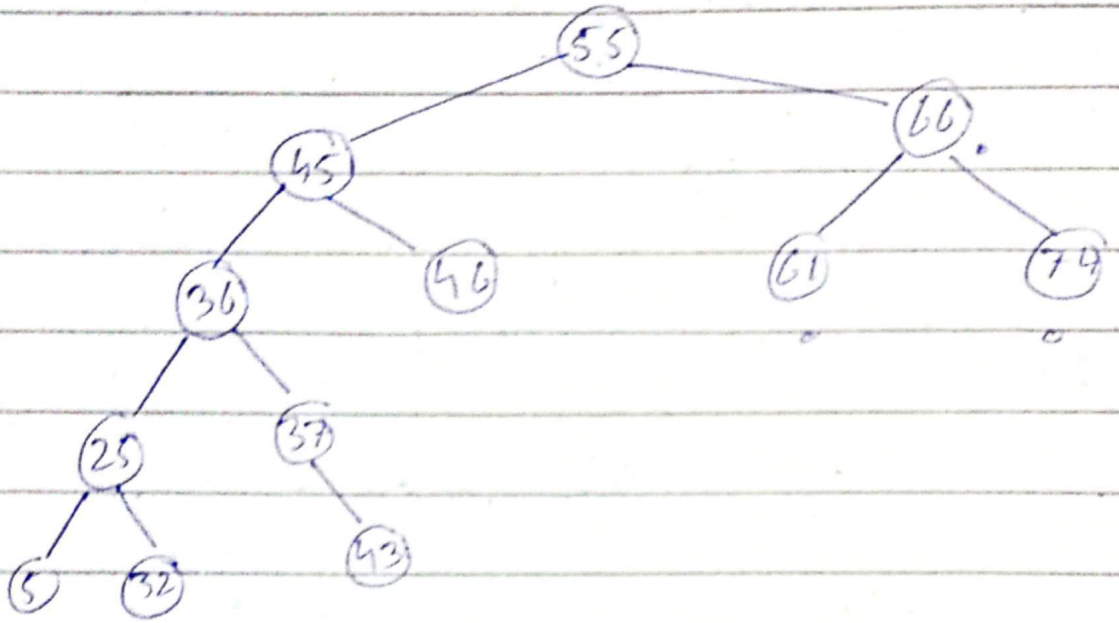


LL :-

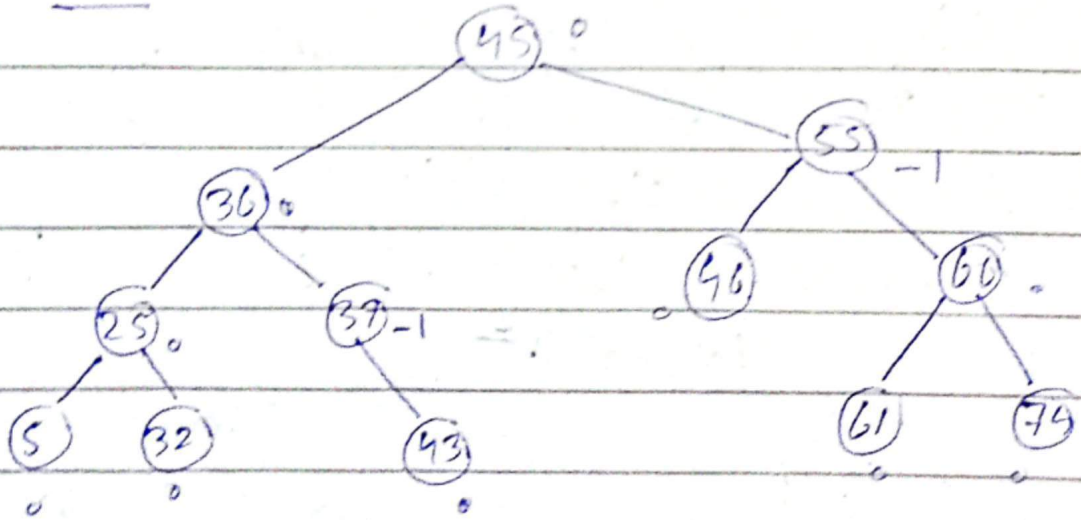


As deletion shrinks the height, so new new pivot is root itself:-

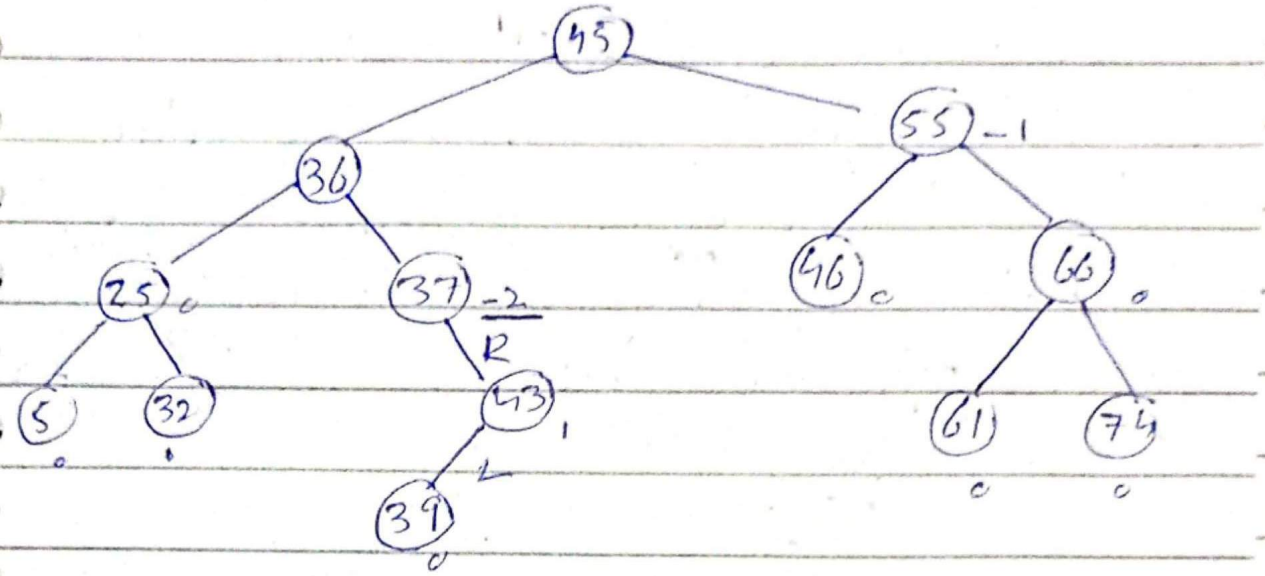
LR. $\xrightarrow{rot}$ LL $\xrightarrow{rot}$ balanced



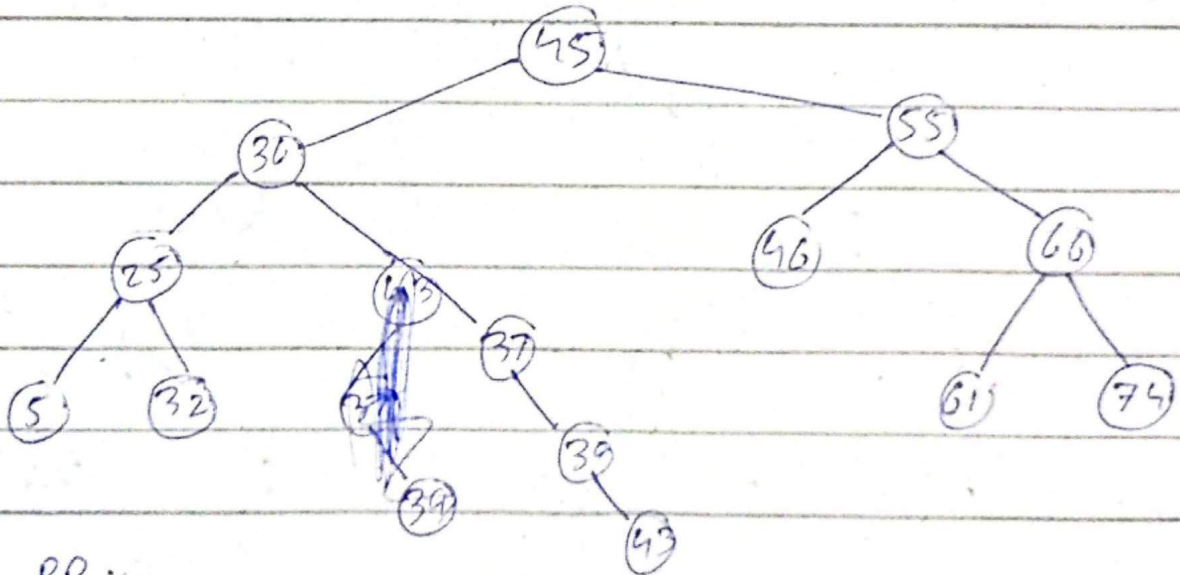
LL :-



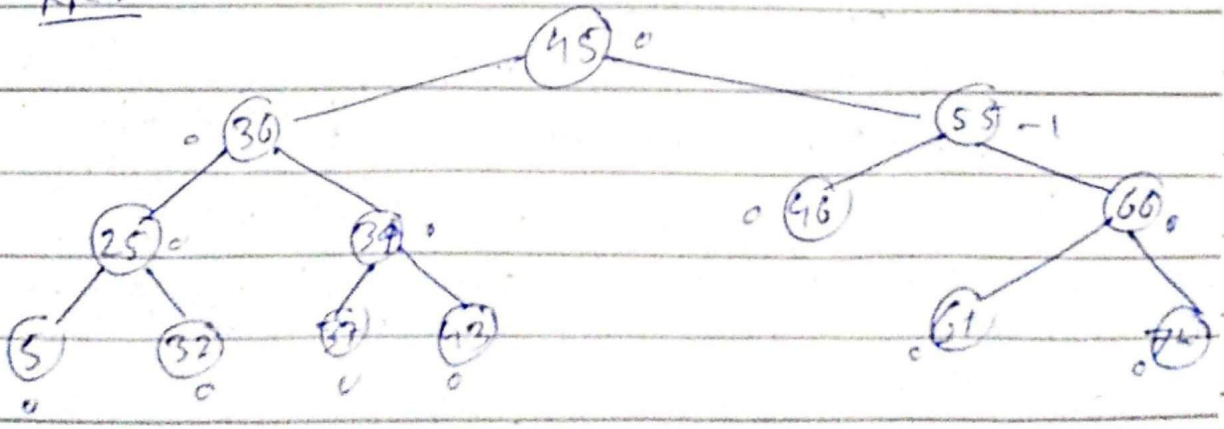
iv- insert 39 :-



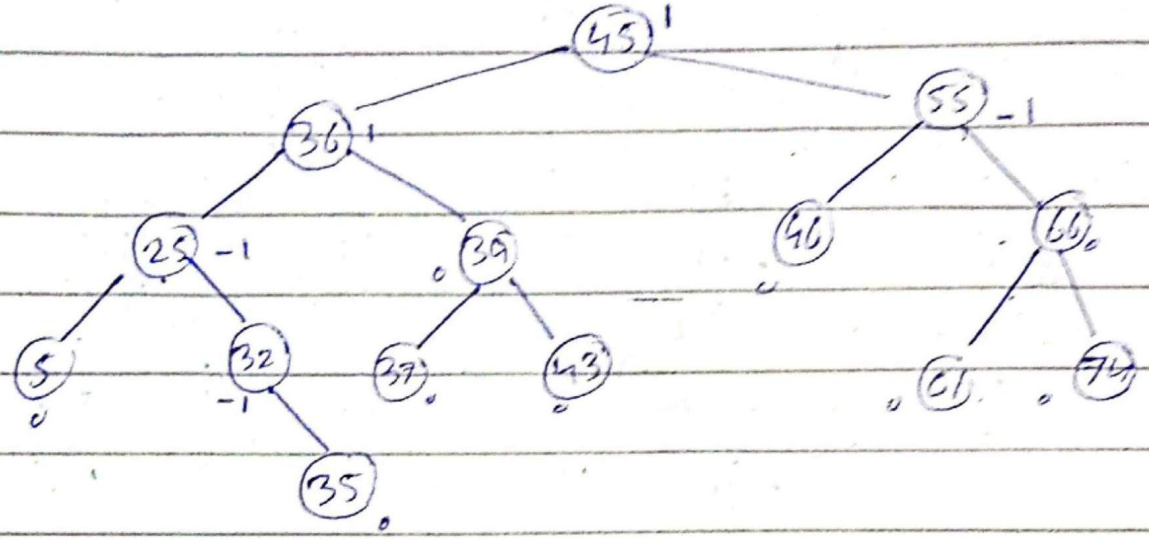
RL $\xrightarrow{rot}$ RR $\xrightarrow{rot}$ balanced



RR :-

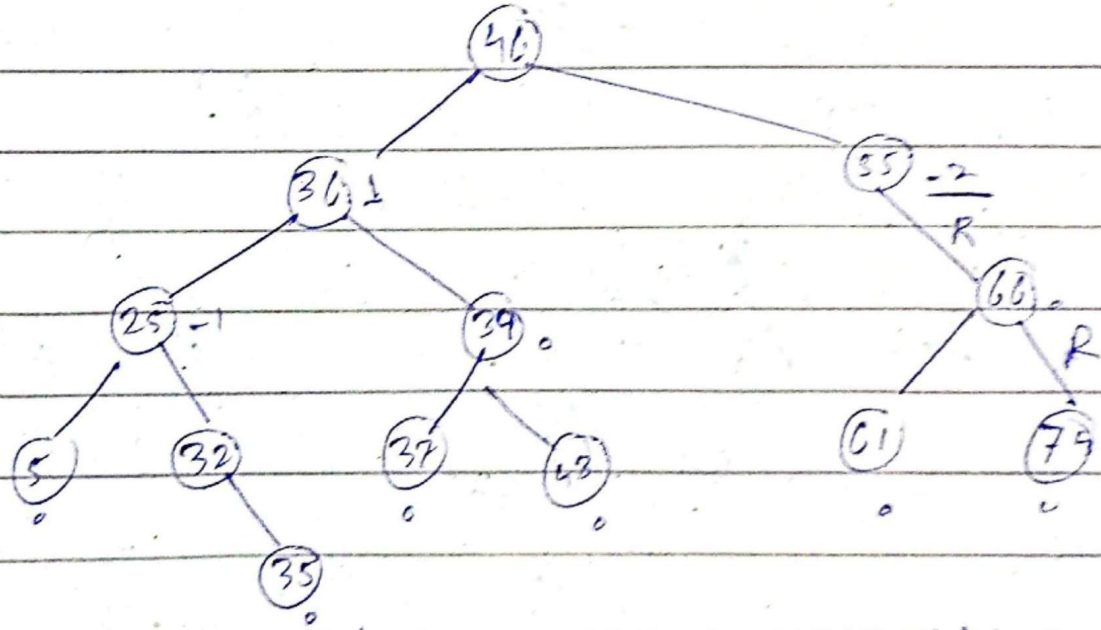


v- insert 35 :-

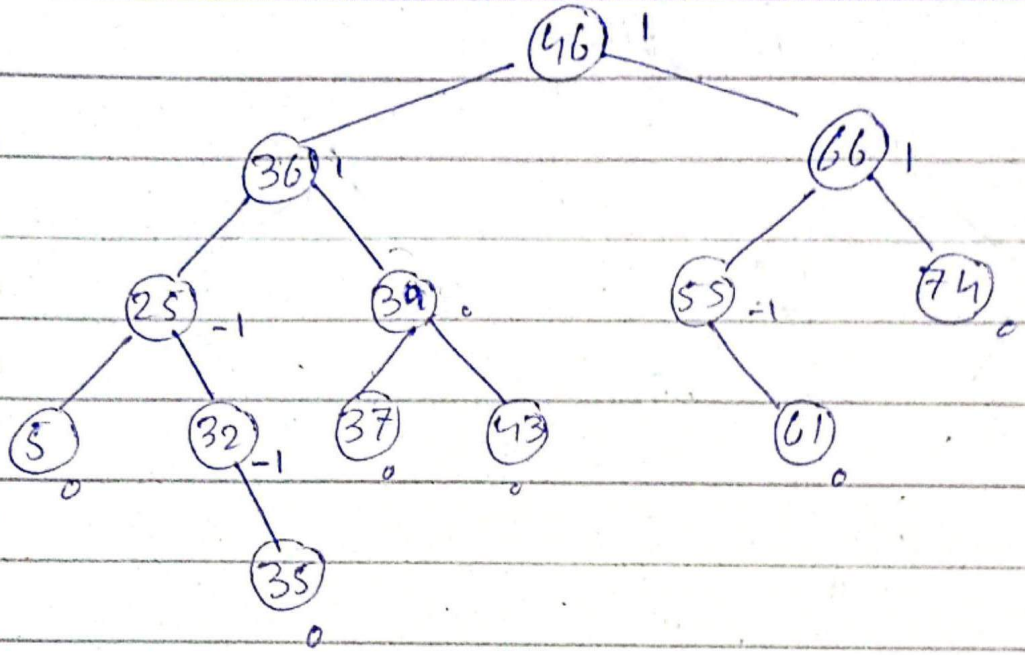


already balanced

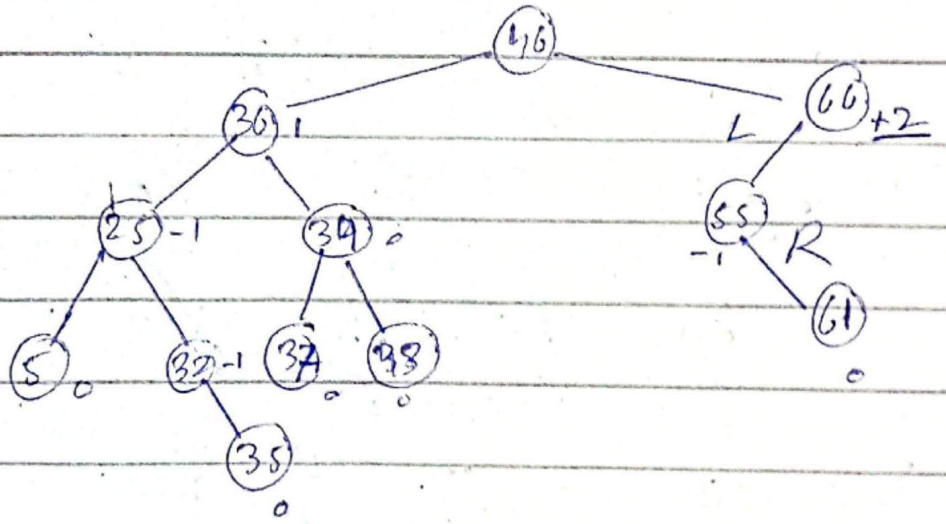
vi- delete 45 :-



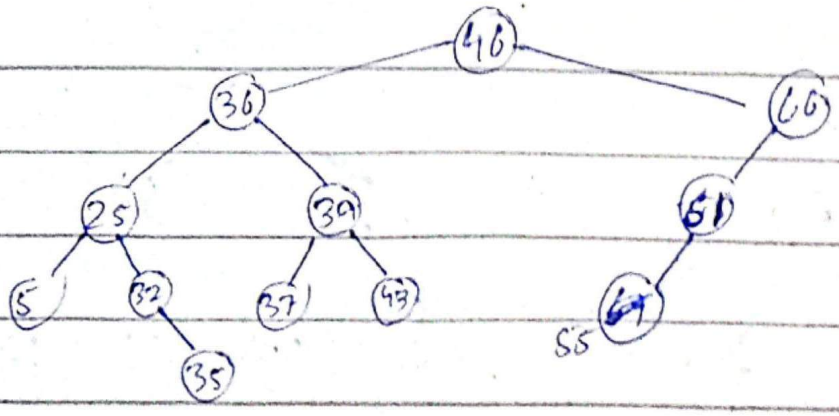
RR → rot balanced (but deletion can cause an imbalance again)

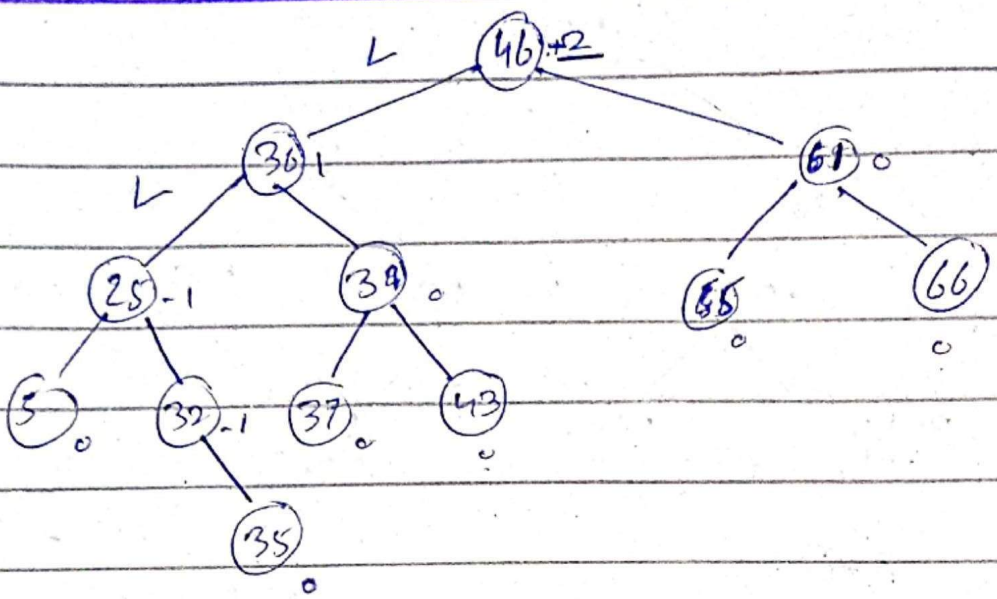


vii- delete 74:-

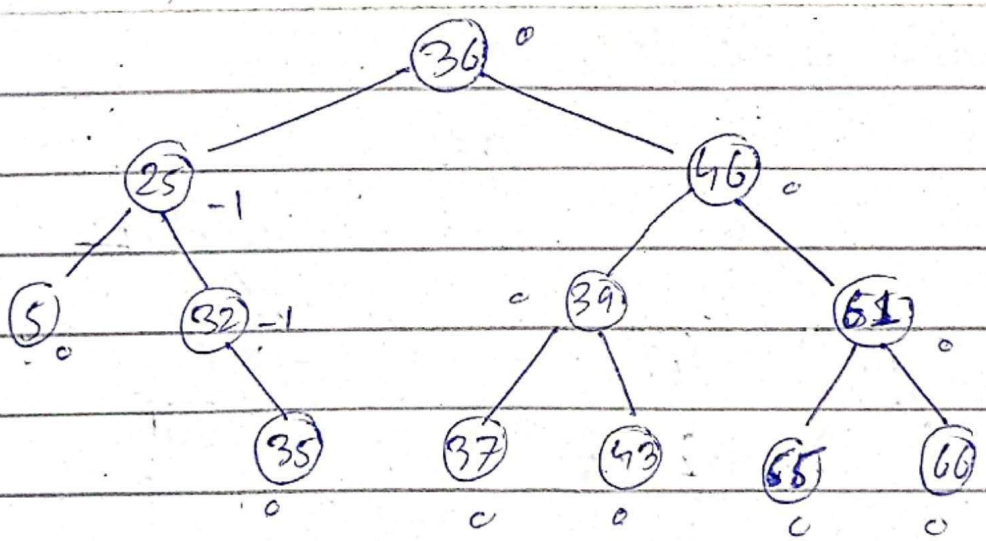


LR $\xrightarrow{rot}$ LL $\xrightarrow{rot}$ balanced

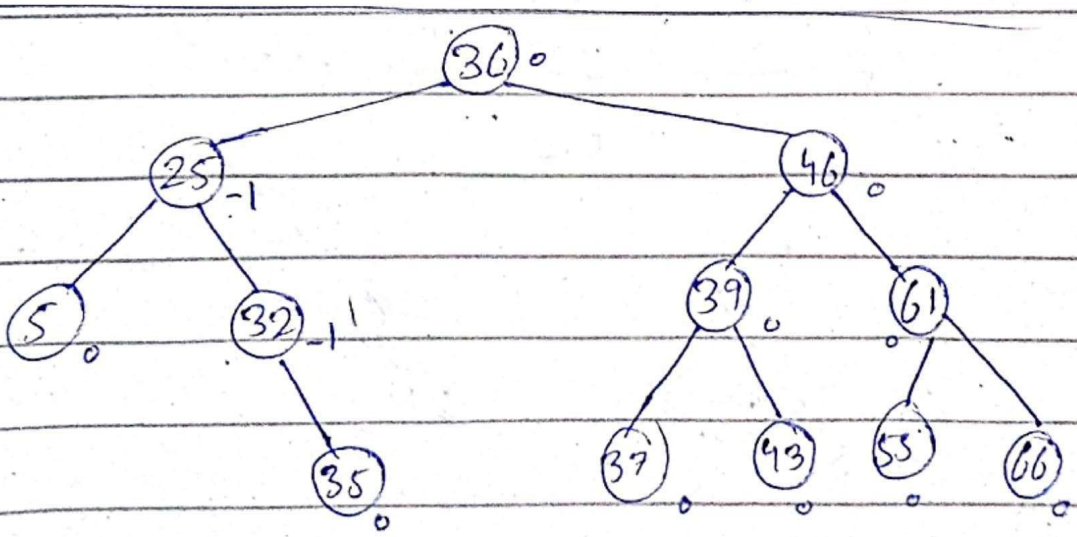




LL :-



Final tree after all (7) operations :-



Pre-order traversal is ~~Left -> Node -> Ri~~

Node -> Left -> Right

36 -> 25 -> 5 -> 32 ->
35 -> 46 -> 39 -> 37 ->
43 -> 61 -> 55 -> 66

Height -> conventionally, leaf is counted 1 (in code) s.o, according to it -> (4) is height
otherwise, if leaf is consided to be edge (c), then (3) is the height.

CamScanner